The documentation and process conversion measures necessary to comply with this revision shall be completed by 31 October 2006.

INCH-POUND

MIL-PRF-19500/370F
31 July 2006
SUPERSEDING
MIL-PRF-19500/370E
5 November 2003

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER,
TYPE 2N3442, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, high-power transistor. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-3).

* 1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

P_T (1) T_A = +25°C	P_T (2) T_C = +25°C	$R_{\theta JC}$ (3)	V_{CBO}	V_{CEO}	V_{EBO}	V_{CER}	I_B	I_C	T_J and T_{STG}
W	W	°C/W	V dc	V dc	V dc	V dc	A dc	A dc	°C
6.0	117	1.5	160	140	7	150	7	10	-65 to +200

(1) Derate linearly 34.2 mW/°C above T_A = +25°C.
(2) See figure 2 for temperature-power derating curves.
(3) See figure 3 for transient thermal impedance graph.

1.4 Primary electrical characteristics.

Limit	h_{FE1} (1) V_{CE} = 4 V dc I_C = 3 A dc	$V_{CE(sat)}$ (1) I_C = 3 A dc I_B = 300 mA dc	h_{fe} V_{CE} = 4 V dc I_C = 3 A dc, f = 100 kHz
		V dc	
Min	20		1.0
Max	70	1.0	

(1) Pulsed (see 4.5.1).

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

AMSC N/A

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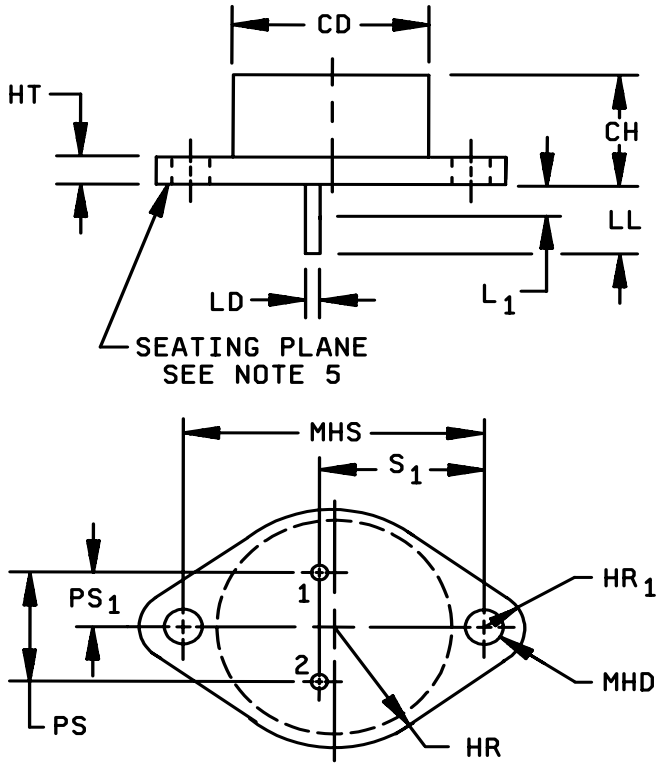


FIGURE 1. Physical dimensions (similar to TO-3).

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.22	
CH	.270	.350	6.86	8.89	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	
HT	.060	.135	1.52	3.43	
LD	.038	.043	0.97	1.09	
LL	.312	.500	7.92	12.70	
L_1		.050		1.27	
MHD	.151	.165	3.84	4.19	
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	4,5
PS_1	.205	.225	5.21	5.72	4,5
S_1	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Lead 1 is emitter; lead 2 is base; and case is collector.
4. Measured at points .050 inch (1.27 mm) - .055 inch (1.4 mm) below the seating plane. When gauge is not used, measurement will be made at the seating plane.
5. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
6. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-3) - Continued.

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

* 3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement
	JANTX, JANTXV levels only
(1) 3c	Thermal impedance (transient), see 4.3.2.
11	h_{FE1}, I_{CEX}
12	See 4.3.1
13	Subgroup 2 of table I herein. ΔI_{CEX} = 100 percent or 500 μA dc whichever is greater; Δh_{FE1} = 25 percent of initial value.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_J = +187.5°C ±12.5°C; V_{CB} = 10 - 30 V dc; $T_A \leq$ +35°C.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table III, group E, subgroup 4 herein.

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* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of table I, subgroup 1 and 2 inspection only (table E-VIB, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Conditions
B3	1037	For solder die attach: 2,000 cycles; V_{CB} = 10 - 30 V dc, $T_A \leq +35°C$.
	or	
	1026	For eutectic die attach: V_{CB} = 10 - 30 V dc, $T_A \leq +35°C$, adjust P_T to achieve $T_J = +175°C$ minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

	Subgroup	Method	Conditions
	C2	2036	Test condition A, weight = 10 pounds ±10 oz., t = 15 s.
*	C5	3131	See 4.3.2, $R_{\theta JC}$ = 1.5°C/W
	C6	1037	For solder die attach: 6,000 cycles; V_{CB} = 10 - 30 V dc, $T_A \leq +35°C$.
		or	
		1026	For eutectic die attach: V_{CB} = 10 - 30 V dc, $T_A \leq +35°C$, adjust P_T to achieve $T_J = +175°C$ minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
	Subgroup 1						
	Visual and mechanical examination	2071					
	Subgroup 2						
	Thermal impedance	3131	See 4.3.2	$Z_{\theta JC}$			°C/W
	Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 3 A dc, pulsed (see 4.5.1), see figure 4	$V_{(BR)CEO}$	140		V dc
	Collector to emitter breakdown voltage	3011	Bias condition B; I_C = 1.5 A dc, R_{BE} = 100 Ω, see figure 4	$V_{(BR)CER}$	150		V dc
	Collector to emitter breakdown voltage	3011	Bias condition A; I_C = 1.5 A dc, V_{EB} = 1.5 V dc, see figure 4	$V_{(BR)CEX}$	160		V dc
	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 125 V dc, V_{EB} = 1.5 V dc	I_{CEX}		0.1	mA dc
*	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 140 V dc	I_{CBO1}		0.1	mA dc
*	Emitter to base cutoff current	3061	Condition D; V_{EB} = 7 V dc	I_{EBO}		1	mA dc
	Base emitter voltage (nonsaturated)	3066	Test condition B; pulsed (see 4.5.1), I_C = 3 A dc, V_{CE} = 4.0 V dc	V_{BE}		1.7	V dc
	Saturation voltage and resistance	3071	Pulsed (see 4.5.1), I_C = 3 A dc, I_B = 300 mA dc	$V_{CE(sat)}$		1.0	V dc
	Forward-current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 3 A dc, pulsed (see 4.5.1)	h_{FE1}	20	70	

See footnote at end of table.

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* TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 140 V dc	I_{CBO2}		1.0	mA dc
Low temperature operation		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 3 A dc, pulsed (see 4.5.1)	h_{FE2}	15		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc, I_C = 3 A dc, f = 100 kHz	h_{fe}	1		
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t = 1 s, see figure 5.				
Test 1		V_{CE} = 11.7 V dc, I_C = 10 A dc				
Test 2		V_{CE} = 78 V dc, I_C = 1.5 A dc				
Test 3		V_{CE} = 125 V dc, I_C = 0.5 A dc				
Electrical measurements		See table I, subgroup 2 herein				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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* TABLE II. Groups B, C, and E delta measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 125 V dc V_{EB} = 1.5 V dc	ΔI_{CEX} 4/	100 percent of initial reading or 500 μA dc, whichever is greater.		

1/ The delta measurements for table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table E-II herein, step 1.
   b. Subgroup 6, see table II herein, step 1.

2/ The delta measurements for table E-VII of MIL-PRF-19500 is subgroup 6, see table II herein, step 1.

3/ The delta measurements for table E-IX of MIL-PRF-19500 are subgroups 1 and 2, see table II herein, step 1.

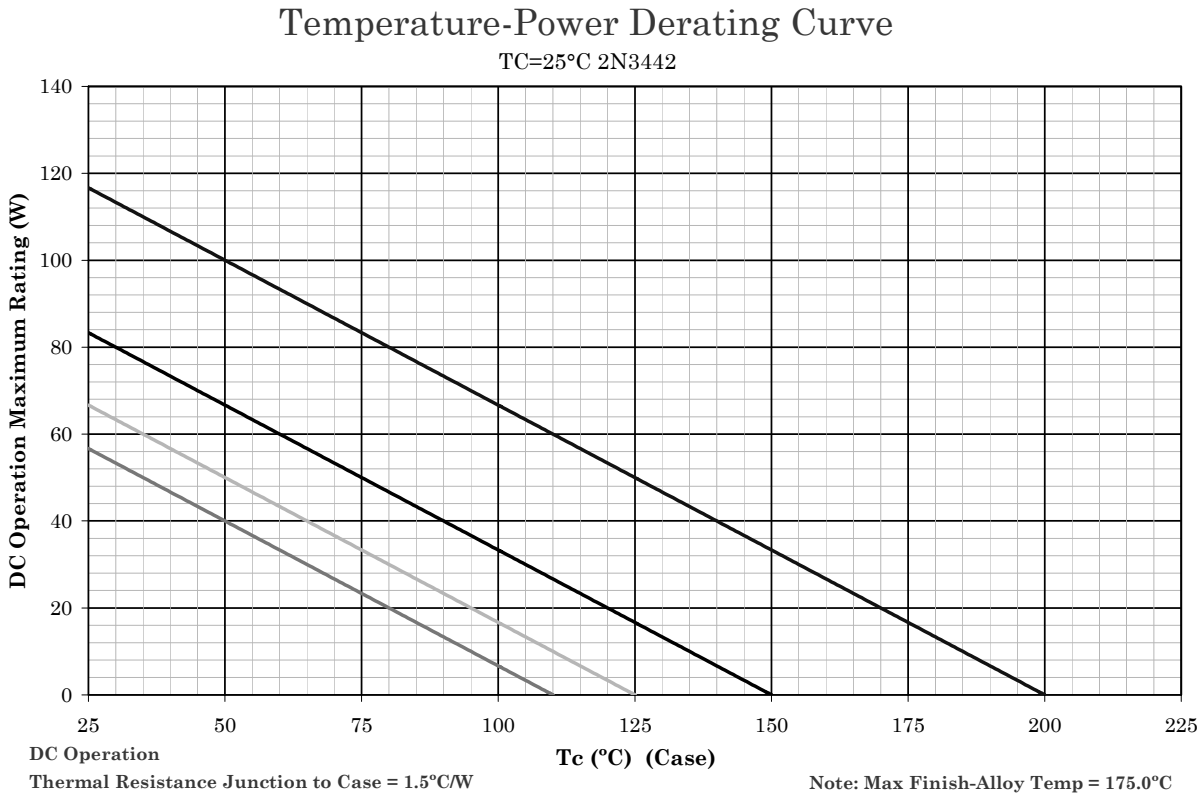
4/ Devices which exceed the table I limits for this test shall not be accepted.

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* TABLE III. Group E inspection (all quality levels) - for qualification and re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	Test condition G, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and table II herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C; V_{CB} = 80 percent of rated voltage; T = 1,000 hours.	
Electrical measurements		See table I, subgroup 2 and table II herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 6			3 devices
ESD	1020	Testing not required for class 3 listing. Testing is required for nonsensitive listing to prove capability.	
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A.	

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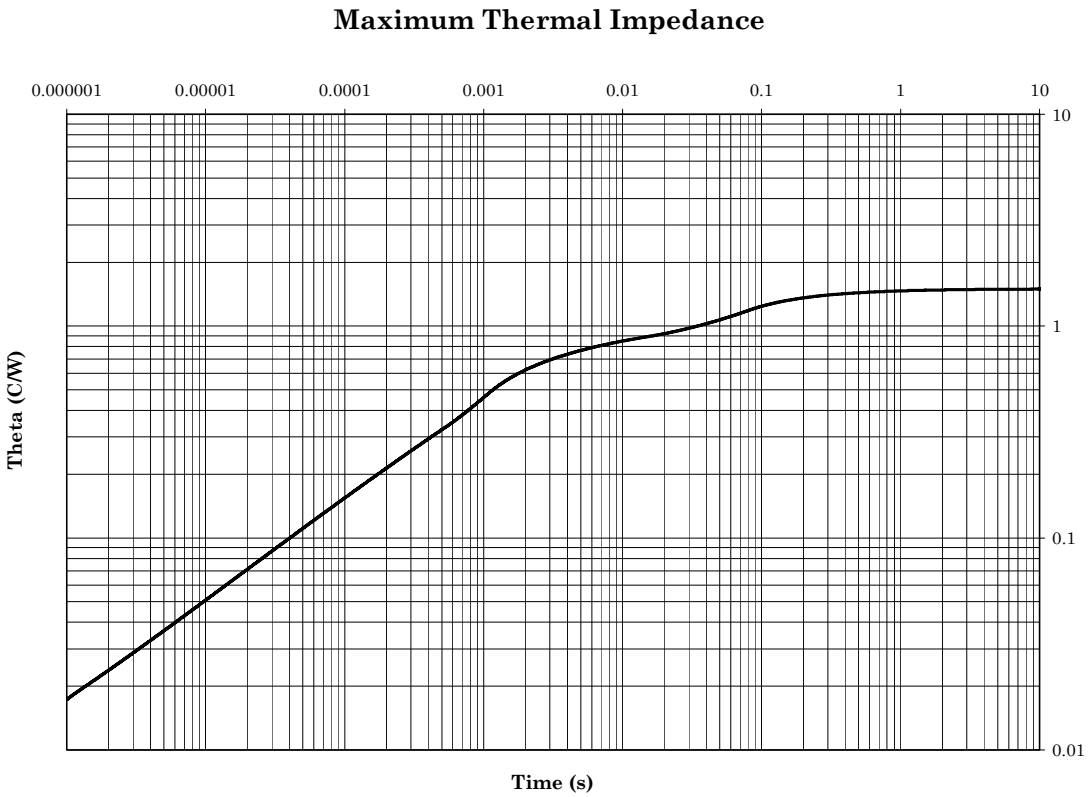


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 2. Temperature derating graph (2N3442, TO-3).

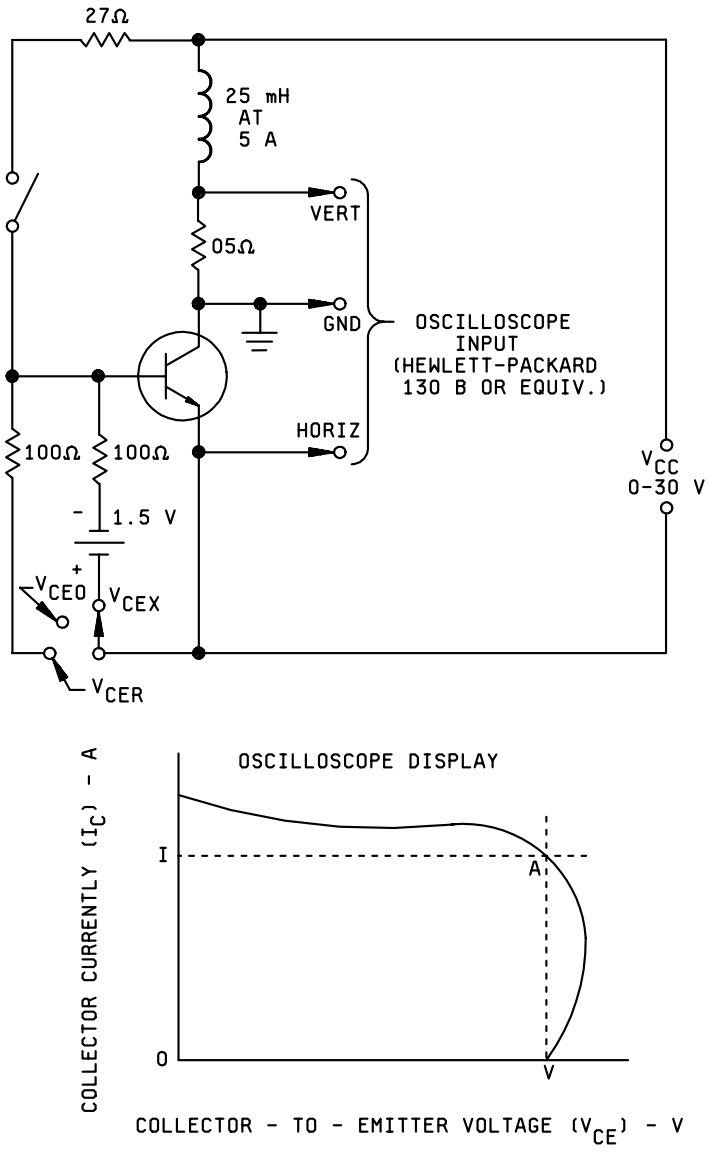
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T_C = +25C. Thermal resistance = 1.5°C/W.

FIGURE 3. Transient thermal impedance graph (2N3442, TO-3).

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NOTE: The voltages V_{CEO}, V_{CER}, or V_{CEX} are acceptable when the trace falls to the right and above point "A".

FIGURE 4. Test circuit for V_{CEO}, V_{CER}, and V_{CEX}.

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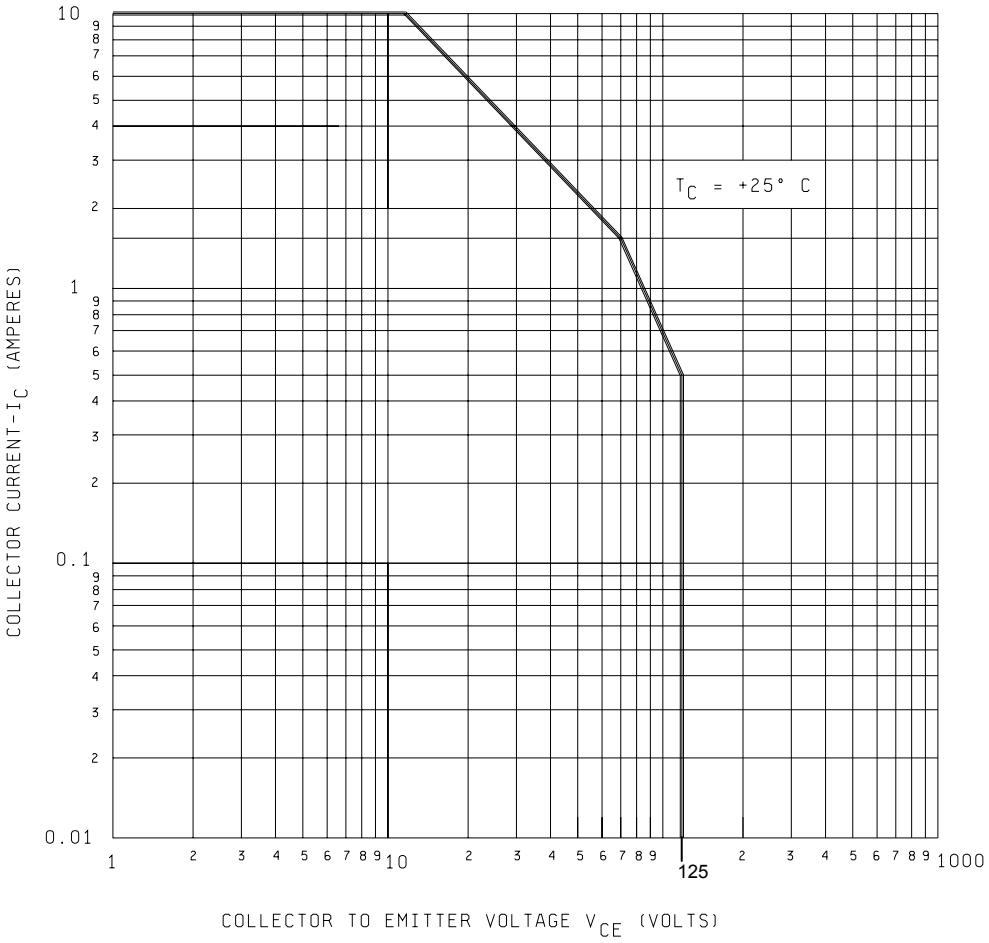


FIGURE 5. Maximum safe operating graph (dc).

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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